Creating directory for docker task 3:

```
[root@ip-172-31-35-127 ~]# mkdir docker-test3
[root@ip-172-31-35-127 ~]# ls -ltrh
total 0
drwxr-xr-x. 2 root root 42 Apr 10 13:30 docker-test
drwxr-xr-x. 3 root root 52 Apr 13 08:43 vcs-task
drwxr-xr-x. 3 root root 70 Apr 15 13:48 microservice-test
drwxr-xr-x. 3 root root 97 Apr 17 15:22 docker-test1
drwxr-xr-x. 2 root root  6 Apr 19 06:31 docker-test3
[root@ip-172-31-35-127 ~]#
```

Creating HTML file:

```
[root@ip-172-31-35-127 docker-test3]# vim index.html
[root@ip-172-31-35-127 docker-test3]#
```

```
root@ip-172-31-35-127:~/doc

Welcome to NGINX page.
this webpage created from custom-nginx.
~
```

Creating Custom-Dockerfile for Nginx:

```
[root@ip-172-31-35-127 docker-test3]# vim Dockerfile
```

```
root@ip-172-31-35-127:~/doc

FROM nginx:latest
COPY ./index.html /usr/share/nginx/html/index/html
EXPOSE 80
```

Build an Image from Dockerfile:

```
[root@ip-172-31-35-127 docker-test3]# docker build -t custom-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/3 : FROM nginx:latest
latest: Pulling from library/nginx
5435b2dcdf5c: Already exists
054715a6bffa: Already exists
88d1d984b765: Already exists
4a038fd18db1: Already exists
84e114c2bb36: Already exists
7b5d674621c2: Already exists
448ea5cac5d5: Already exists
Digest: sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
 ---> a716c9c12c38
Step 2/3 : COPY ./index.html /usr/share/nginx/html/index/html
 ---> 98d12fe73522
Step 3/3 : EXPOSE 80
 ---> Running in 3a8668068d26
 ---> Removed intermediate container 3a8668068d26
 ---> 3c6ca1abac42
Successfully built 3c6ca1abac42
Successfully tagged custom-nginx:latest
[root@ip-172-31-35-127 docker-test3]#
```

Docker image created:

```
[root@ip-172-31-35-127 docker-test3]# docker images
REPOSITORY                TAG       IMAGE ID       CREATED          SIZE
custom-nginx              latest    3c6ca1abac42   31 seconds ago   161MB
docker-test1-my-website   latest    fc7ede8f3095   40 hours ago     62.2MB
nginx                     alpine    812d47f806db   3 days ago       62.2MB
nareshvj04/naresh         myapp     290ab06a1fe5   8 days ago       161MB
nodejs                    v1        9274d1af29a1   8 days ago       62.2MB
nginx                     latest    a716c9c12c38   11 days ago      161MB
httpd                     latest    e48c16e072a6   12 days ago      117MB
[root@ip-172-31-35-127 docker-test3]#
```

Change tag according to DockerHub username to push Docker image:

```
[root@ip-172-31-35-127 docker-test3]# docker tag custom-nginx:latest nareshvj04/custom-nginx:v1
[root@ip-172-31-35-127 docker-test3]# docker images
REPOSITORY                TAG       IMAGE ID       CREATED          SIZE
nareshvj04/custom-nginx   v1        3c6ca1abac42   6 minutes ago    161MB
custom-nginx              latest    3c6ca1abac42   6 minutes ago    161MB
docker-test1-my-website   latest    fc7ede8f3095   40 hours ago     62.2MB
nginx                     alpine    812d47f806db   3 days ago       62.2MB
nareshvj04/naresh         myapp     290ab06a1fe5   8 days ago       161MB
nodejs                    v1        9274d1af29a1   8 days ago       62.2MB
nginx                     latest    a716c9c12c38   11 days ago      161MB
httpd                     latest    e48c16e072a6   12 days ago      117MB
```

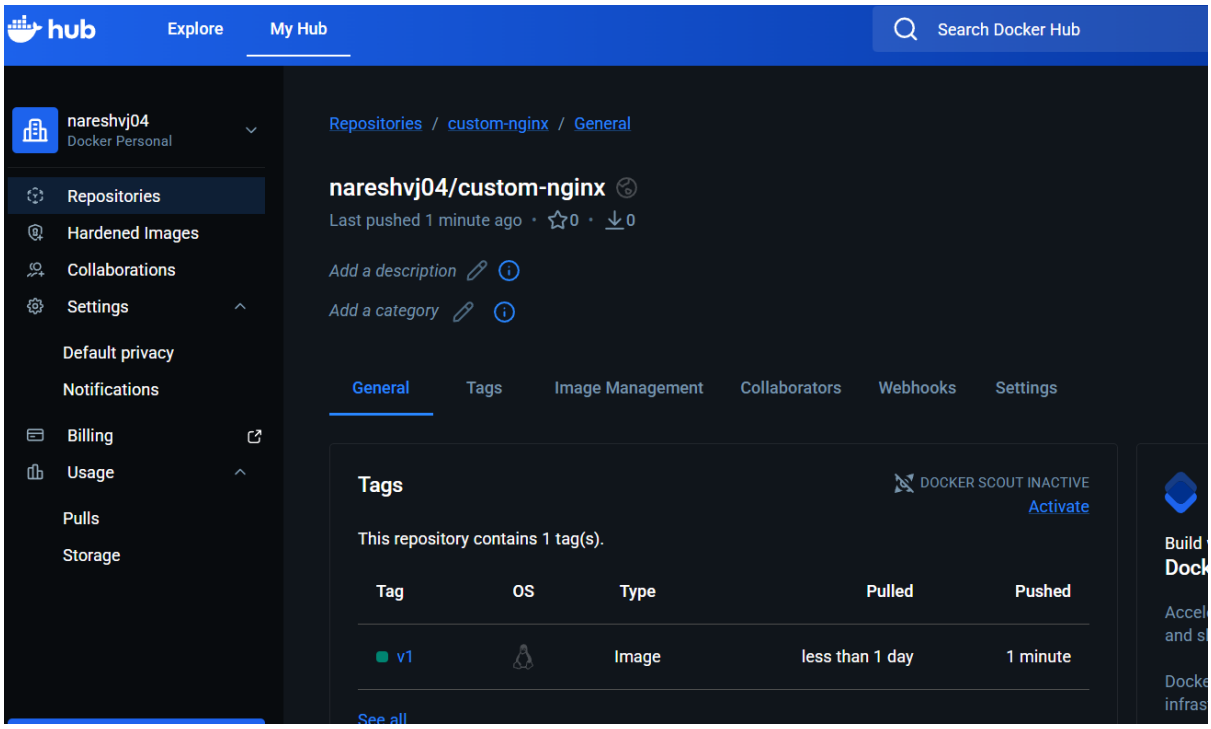
Login to DockerHub:

```
[root@ip-172-31-35-127 docker-test3]# docker login
Authenticating with existing credentials...
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-35-127 docker-test3]#
```

Image pushed to DockerHub:

```
[root@ip-172-31-35-127 docker-test3]# docker push nareshvj04/custom-nginx:v1
The push refers to repository [docker.io/nareshvj04/custom-nginx]
a617c5dfea9d: Pushed
9d09dfdcffeb: Mounted from nareshvj04/naresh
5ca2f881e5bf: Mounted from nareshvj04/naresh
db010d6b6a3c: Mounted from nareshvj04/naresh
3fd3ceafa064: Mounted from nareshvj04/naresh
4ef7ac306afb: Mounted from nareshvj04/naresh
a4b536aa4bf8: Mounted from nareshvj04/naresh
60e70dddd9ea: Mounted from nareshvj04/naresh
v1: digest: sha256:0bb709d3710be2856c175157f62f2f66ff37fc3c481cc78bc9d477d06ab161a4 size: 1986
[root@ip-172-31-35-127 docker-test3]#
```



Create directory for volume bind:

```
[root@ip-172-31-35-127 docker-test3]# mkdir my-data
[root@ip-172-31-35-127 docker-test3]# ls -ltrh
total 12K
```

Creating Docker-Compose file, binding host volume to specific directory:

```
services:
  custom-nginx:
    image: nareshvj04/custom-nginx:v1
    build: .
    ports:
      - "8080:80"
    volumes:
      - ./my-data:/var/opt/nginx
```

```
[root@ip-172-31-35-127 docker-test3]# mkdir my-data
[root@ip-172-31-35-127 docker-test3]# ls -ltrh
total 12K
-rw-r--r--. 1 root root  63 Apr 19 06:37 index.html
-rw-r--r--. 1 root root  79 Apr 19 06:54 Dockerfile
-rw-r--r--. 1 root root 154 Apr 19 07:16 docker-compose.yml
drwxr-xr-x. 2 root root   6 Apr 19 07:16 my-data
```

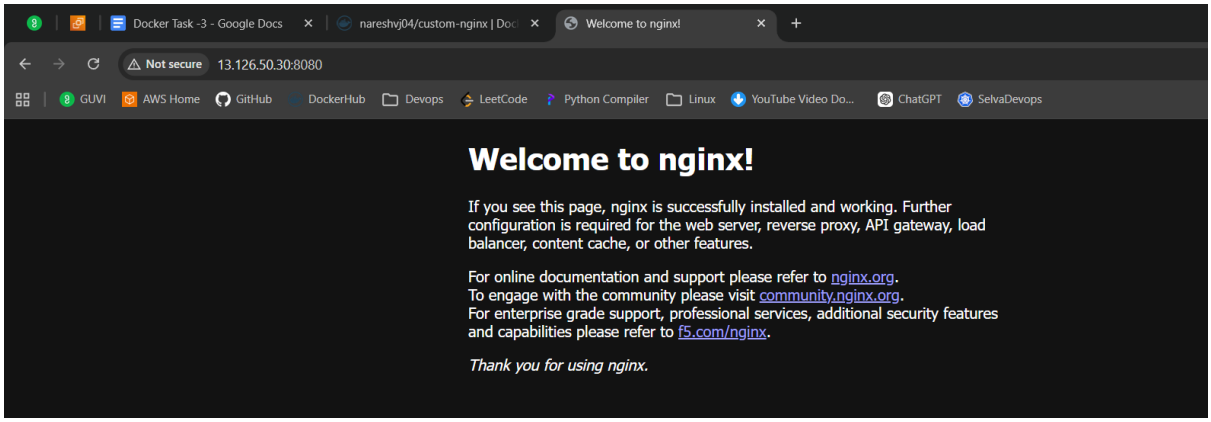
Deploying Custom-Nginx using docker compose:

```
[root@ip-172-31-35-127 docker-test3]# docker compose up -d
[+] up 2/2
 ✔Network docker-test3_default          Created                                    0.1s
 ✔Container docker-test3-custom-nginx-1 Started                                    0.5s
[root@ip-172-31-35-127 docker-test3]#
```

Container created and working :

```
[root@ip-172-31-35-127 docker-test3]# docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED              STATUS              PORTS
                                   NAMES
3d9f2e70e73c   nareshvj04/custom-nginx:v1   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:80
80->80/tcp, :::8080->80/tcp   docker-test3-custom-nginx-1
[root@ip-172-31-35-127 docker-test3]#
```

Webpage working :



Login to the container:

```
[root@ip-172-31-35-127 docker-test3]# docker exec -it 3d9f2e70e73c /bin/bash
root@3d9f2e70e73c:/#
```

Creating file to the container:

```
[root@ip-172-31-35-127 docker-test3]# docker exec -it 3d9f2e70e73c /bin/bash
root@3d9f2e70e73c:/# cd /var/opt/nginx
root@3d9f2e70e73c:/var/opt/nginx# touch testfile.txt
root@3d9f2e70e73c:/var/opt/nginx#
```

Exit from Container, checking the file from the host machine if file created or not:

```
root@3d9f2e70e73c:/usr/share/nginx/html#
exit
[root@ip-172-31-35-127 docker-test3]# ls -ltrh
total 12K
-rw-r--r--. 1 root root  63 Apr 19 06:37 index.html
-rw-r--r--. 1 root root  79 Apr 19 06:54 Dockerfile
-rw-r--r--. 1 root root 154 Apr 19 07:16 docker-compose.yml
drwxr-xr-x. 2 root root  26 Apr 19 07:26 my-data
[root@ip-172-31-35-127 docker-test3]# cd my-data/
[root@ip-172-31-35-127 my-data]# ls -ltrh
total 0
-rw-r--r--. 1 root root 0 Apr 19 07:26 testfile.txt
[root@ip-172-31-35-127 my-data]#
```

File created, so volume bind host machine to the Created Container:

Note:

Index.html move to wrong path, so Index.html page is not working, but Nginx default page is loaded successfully.